DriveHub Motors – Part 3 Improvement Proposal

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1. Introduction

Part 1 delivered the planning and design of the DriveHub Motors website, and Part 2 delivered the six linked pages in HTML and CSS. Part 3 adds JavaScript so that the website responds to visitors instead of only displaying content. This proposal explains what was added, what improved compared with the earlier version, and the limits of the current build.

2. Goals of Part 3

- Make the site interactive so visitors can find vehicles faster.
- Check form input before it is submitted, so that fewer incomplete or incorrect enquiries are sent.
- Give visitors instant feedback after they contact the dealership or make an enquiry.
- Improve the look and feel of the pages without slowing them down.
- Replace the static map with an interactive one.

These goals support the original website objectives: showcase current inventory, build trust and credibility, and generate qualified enquiries and test-drive bookings.

3. Summary of What Improved

Page	Before (Part 2)	After (Part 3)
All pages	Content appeared all at once and the page felt static.	Sections fade in smoothly as the visitor scrolls (Intersection Observer). Reduced-motion settings are respected.

Page	Before (Part 2)	After (Part 3)
Products and Services	Vehicles were fixed photos grouped under three headings.	A filter bar (All, SUVs, Sports, Family) builds the vehicle list dynamically and updates the category description.
Products and Services	No way to ask about a vehicle on the page.	New enquiry form with a category and enquiry type. It returns a relevant reply for pricing, financing, trade-in or test drive.
Gallery	Photos were small and could not be enlarged.	Clicking a photo opens a lightbox with a caption, next/previous buttons, arrow-key navigation and Esc to close.
Contact Us	Form relied only on the browser's basic checks and had no message type.	Full JavaScript validation with clear error messages beside each field, a message-type dropdown and a success message with a link to send the email.
Contact Us	Embedded Google Maps iframe.	Interactive Leaflet and OpenStreetMap map with a marker and popup for the showroom in Braamfontein.

4. Detailed Improvements

4.1 Scroll-reveal animation

Main content sections fade and slide into view as the visitor scrolls. This makes the pages feel more polished and guides attention down the page. If a browser does not support the Intersection Observer feature, or the visitor prefers reduced motion, the content simply appears normally.

4.2 Vehicle filter (Products and Services)

The vehicle list is now generated from a JavaScript array of vehicles. Each vehicle has a name, category, image and description. When a visitor clicks a filter button, only the matching vehicles are displayed and the description text changes to match. Adding a new vehicle in future only requires adding one entry to the array rather than editing the page layout.

4.3 Enquiry form (Products and Services)

Visitors choose a vehicle category and the type of enquiry. After validation, the page displays a response tailored to their choice, such as test-drive availability or a financing note, addressed to them by first name. This answers common questions immediately and reduces the effort needed to contact the dealership.

4.4 Gallery lightbox

Gallery photos can be opened in a larger overlay. Visitors can move between photos with on-screen buttons or the left and right arrow keys, and close the overlay with the close button, a click outside the photo, or the Esc key. Captions are taken from the existing figure captions, so no content is duplicated.

4.5 Contact form validation

The form checks that the name has at least two characters, the email address is in a valid format, the phone number (if given) contains only valid characters, a message type is selected, and the message is at least ten characters long. Errors appear in red next to the field that needs fixing, and the field is highlighted. When the form is valid, the visitor sees a confirmation message with a link that opens their email app with the message pre-filled.

4.6 Interactive map

The embedded Google Maps frame was replaced with a Leaflet map using OpenStreetMap tiles. Visitors can zoom and drag the map, and a popup shows the dealership name and address. Both libraries are free to use.

5. Fixes Made While Adding JavaScript

- The pages now load the script from the correct folder (`JAVASCRIPT/script.js`).
- The stylesheet link was corrected to match the real folder name (`CSS/Style.css`), so the site also works on case-sensitive web hosts.
- Image paths used by the script were corrected to the real `IMAGE` folder.
- A broken image tag on the Gallery page (Porsche interior) was repaired.
- The Contact page structure was cleaned up (misplaced body tag removed) so that the HTML is valid.

6. Technical Requirements

- **Languages:** HTML5, CSS3 and vanilla JavaScript (no frameworks).

- **Library:** Leaflet 1.9.4 for the map, loaded from a CDN.
- **Browser features used:** Intersection Observer, DOM manipulation, event listeners and `setTimeout` for a simulated send delay.
- **Files:** `script.js` is shared by all six pages, and each function checks whether its page elements exist before running.

7. Testing Approach

- Click every filter button and confirm the correct vehicles and description appear.
- Submit both forms empty, then with invalid data, then with valid data, and confirm the messages are correct.
- Open the lightbox on every gallery photo and test the buttons, arrow keys, Esc and clicking outside.
- Resize the browser to phone width and confirm the layout still works.
- Open the site with the internet disconnected to confirm the page still loads (only the map tiles are affected).

8. Limitations and Future Improvements

- The site has no backend, so the contact form does not send an email by itself. It prepares an email for the visitor to send. A future version could use a server or a form service.
- The enquiry replies use general wording. They are not real prices or live stock, and a future version could read from a stock database.
- The map needs an internet connection to load its tiles.
- Possible next steps: a search box for vehicles, a vehicle details page, and saving a visitor's favourite vehicles.

9. Budget

No extra cost. All code was written by hand, and Leaflet and OpenStreetMap are free and open source.

10. References

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